

we weather the storms



Mary Tallontire
Hotel Generation 2026



we weather the storms is an exhibition that functions as a multiplayer game experience with audiences connecting to a singular, living virtual world visible through multiple screens and immersive projections on their phones. Collectively, audiences react and respond to an unpredictable, dramatic weather system within a virtual ecosystem inspired by solarpunk aesthetics, with each player having agency as an individual but requiring the collective to make major impacts. Anchored in optimism and hope in an era of climate anxiety, the virtual world is a counter-dystopia.

Whilst *we weather the storms* takes inspiration from solarpunk aesthetics, it acknowledges that we have the means of living in a sustainable 'utopian' future now, though this is withheld from the majority due to capital constraints. Leaving behind aspects of science fiction, instead the exhibition intends to highlight the capitalist, colonial systems that need dismantling for us to achieve a sustainable future, looking at how we can reframe our technology to now push us towards a solarpunk world.

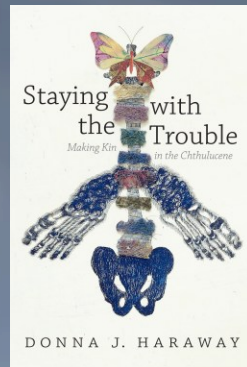
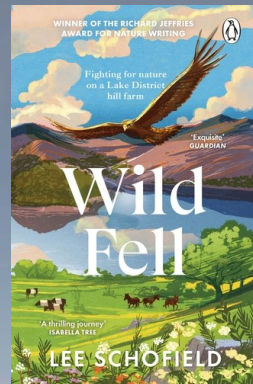
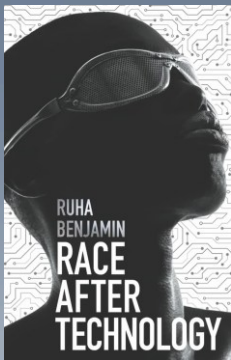
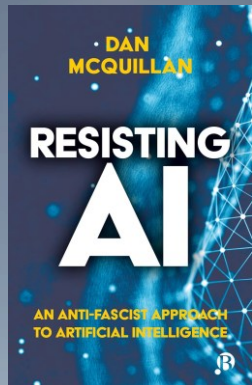
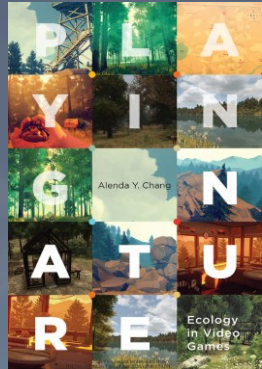
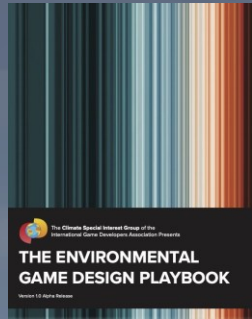
Rooted in the notion that there can be no separation between technology and ecology, the exhibition examines a false sense of immateriality in networking and gaming. It strives for a technological landscape that does not depend on profit-driven technologies that largely end up as e-waste, which in the UK is then rapidly exported to the global south.

The multiplayer controls to engage with the digital ecosystem are accessed through audience members' phones, reclaiming mobile phones as a technology of connection and communication instead of the tools of corporate control fostering dopamine dependence, AI slop and algorithms of surveillance. The exhibition purposefully uses pre-existing technologies that aren't cutting edge, reuses digital tools and algorithms, and is largely open-source, with the multiplayer server running on solar energy on an old laptop in Leeds. The exhibition promotes sustainable innovation with technologies that already exist, without falling into a techno-solutionist mindset, acknowledging that not all problems need to be solved with tech.

In an era of culture wars and division, the exhibition aims to unite audiences through play, using game mechanics to allow audiences to experience and experiment with an alternative system of collective resource management, inspiring eco hope and utilising play as a tool of resistance. It looks to foster connection between the physical bodies that exist in the present between the digital projections and screens, generating a sense of community amongst players, whilst also providing a hopeful, reflective, meditative space away from corporatised mindfulness. The immersive experience offers up a different world and understanding of how our lives could impact ecosystems in a more positive way.

RESEARCH

core readings



"Perhaps it is precisely in the realm of play, outside the dictates of teleology, settled categories, and function, that serious worldliness and recuperation become possible"

- Donna Haraway, Staying with the Trouble

"As we struggle to come to terms with the scale of environmental change and injury, games that make the leap from single-player to multiplayer, that channel the collective, social action of players towards more than just virtual farming or wordplay, may show us the way past apathy and individual limitation"

- Alenda Y. Chang

inspired by artist Kara Stone and her solar powered game server in Canada, I've been working on my server to reduce the carbon footprint of this exhibition



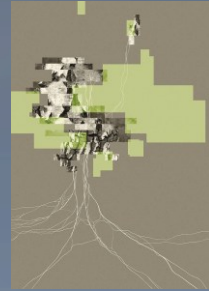
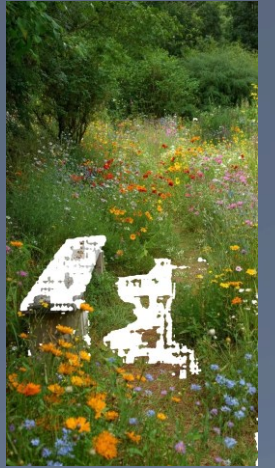
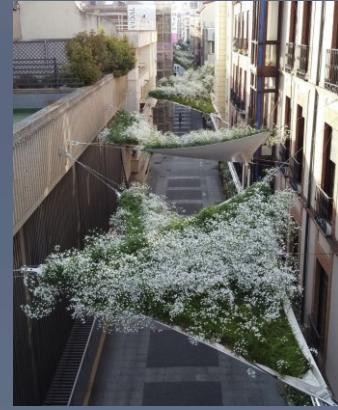
my teenage laptop (I've recently upgraded some parts using second-hand tech)



solar panels on a family member's house in Leeds

= a predominantly solar powered multiplayer game server

MOODBOARD



simple skyboxes and gradients



fonts

Pixel Operator
HandJet Bold
HandJet



AUDIENCE JOURNEY

Soundscape heard throughout, and lighting linked to the ecosystem's 15 minute day-to-night cycle. Built to allow audience members to jump in at any point.

show description

digital forest room

screens, wires, lights and PCs used to display the digital ecosystem build an abstract forest, emphasizing the link between technology and ecology

screens display different perspectives of the digital world, with more camera shots, third person animal walking cameras and in game information such as weather forecast

water projection that reflects the amount of water collected in game.

immersive projection room with projection mapped water shader and stepping stones

legacy screen

touch screen for players leaving to create a small, lasting impact on the digital ecosystem they won't see the effects of.

cannot be seen by players still in the installation, players are not told it is there.

means players in the space are reacting to the impacts of audience members they may have never met, adding another multiplayer aspect even if attending the installation alone.

NFC tag / website link

where audiences connect to the digital ecosystem through a web browser. Each player will be randomly assigned one of three different ways to interact.

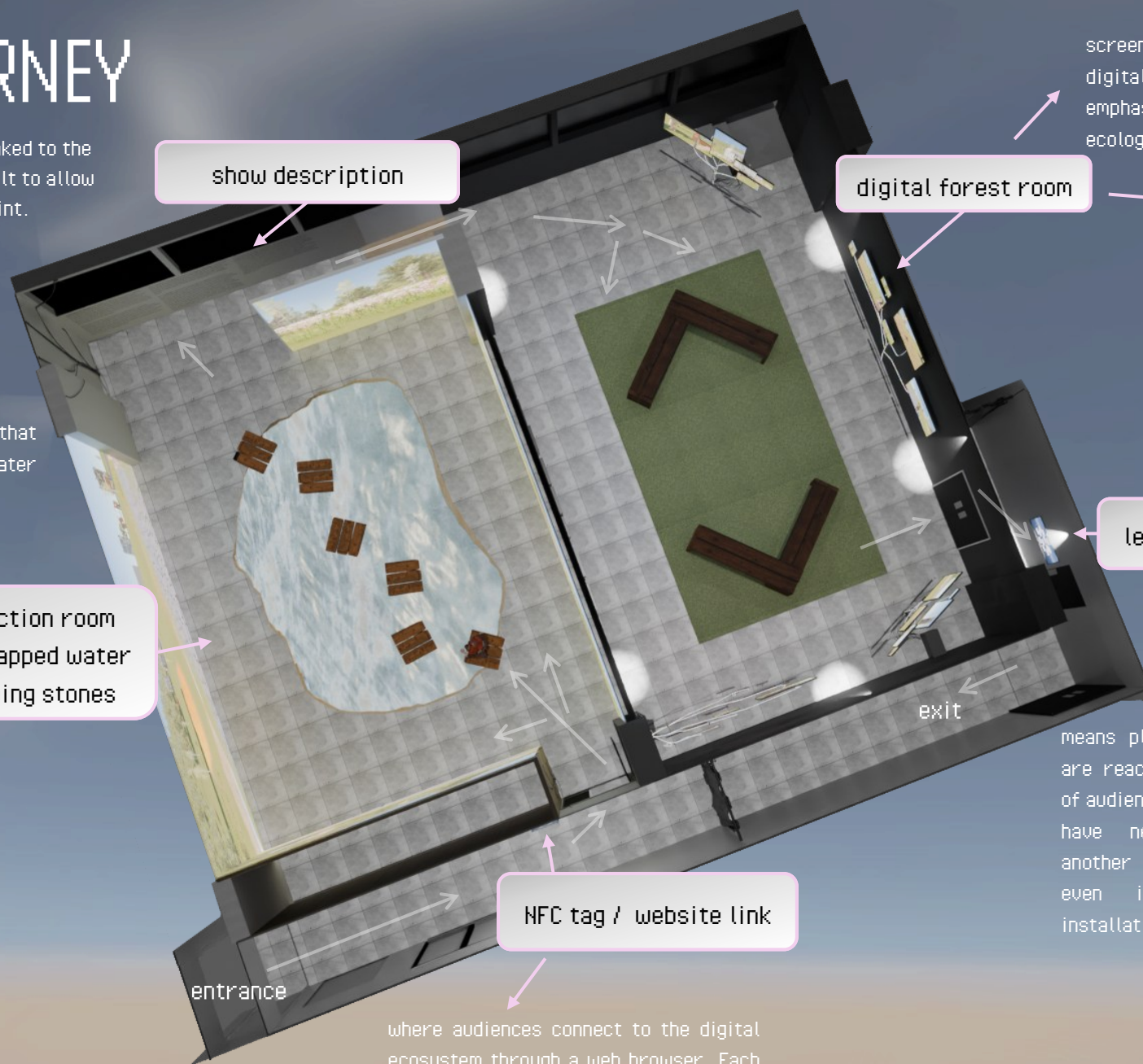
VIRTUAL WALKTHROUGH HERE

window into the digital ecosystem

stepping stones mean the audience immediately encounter a physical, playful interaction. requires them to think about where they are physically in the in relation to other audience members and potentially starts initial conversations.

entrance

exit





projection room



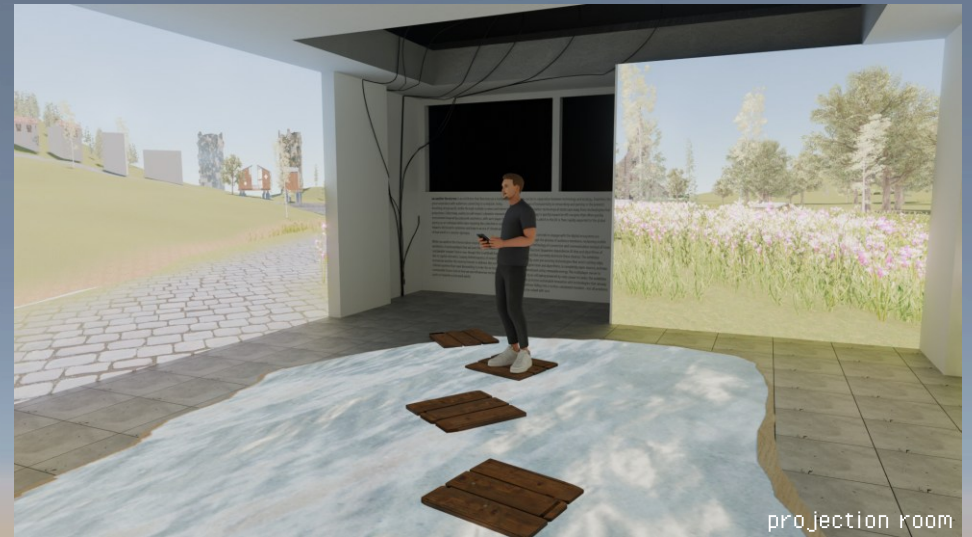
forest room



forest room



legacy screen



projection room

UNITY WIP VIDEO HERE

MULTIPLAYER GAME MECHANICS

Anchored in slow, reflective play, *we weather the storms* aims to build connections between audience members, generating a sense of community amongst participants through a collective resource management game. The interactive space hopes to build trust within audiences in community and collective action, creating moments of eco hope to communicate collective environmental responsibility. Only players physically in the installation will be able to alter the environment.

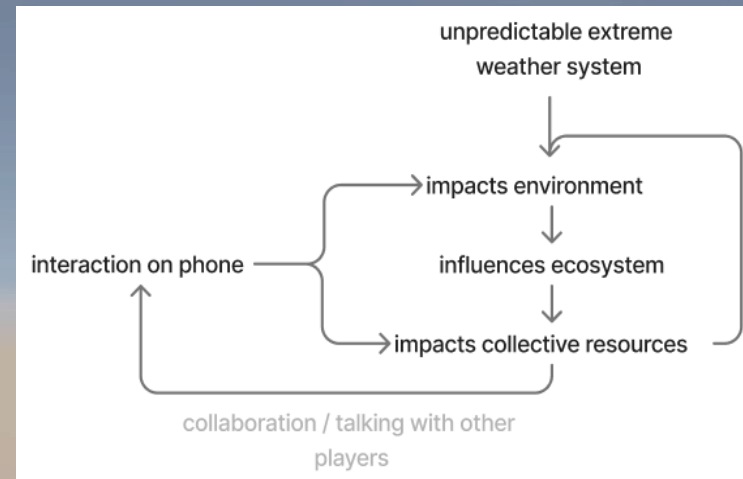
The collective goal of the players is to respond to the irregular weather patterns within the environment, to help the simulated ecosystem and society to recover faster than it would do so naturally. Every audience member can become an active member of the digital ecosystem using their mobile phone as a controller. The controllers take the form of a website that audiences connect to through using the NFC tag on the wall outside the entrance to the exhibition.

When players connect, they are randomly assigned controls, with there being three different interaction methods. This means players must work together to best respond to the irregular weather using all interactions available to them. Upon joining, players become collectively responsible for managing resources within the environment.

The simulated ecosystem will always eventually rebalance itself, removing elements of pressure from gameplay and allowing for a more reflective and hopeful player experience. This means time becomes a mechanic, whilst also allowing the installation to continue to run like normal when there are no or a small number of active players, allowing the ecosystem to be active for passive, non-playing audience members as well.



more about controllers on next page



The legacy screen is the final method of interacting with the game, sitting outside the main installation, at the end of the experience. This screen provides some audio feedback to players; however, audience members will not see the impact of their decision - unless they chose to go back into the installation.

CONTROLLERS

Wireframes are for iPhone 17
first draft UI

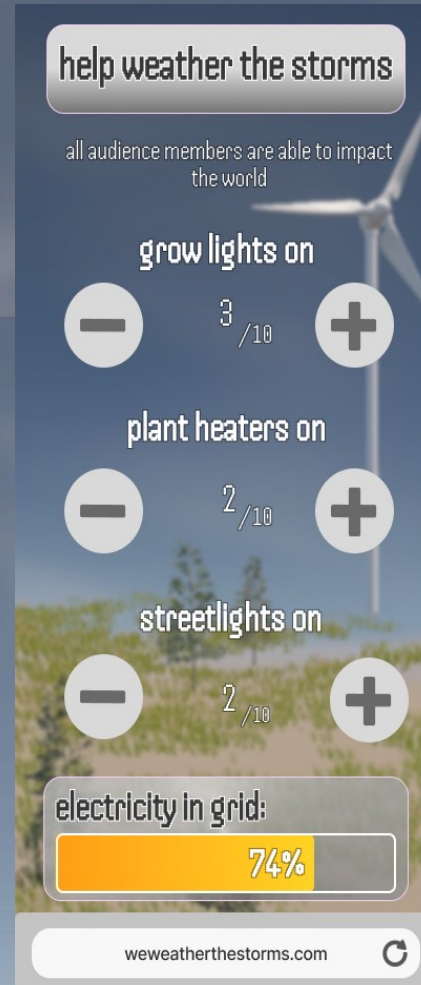
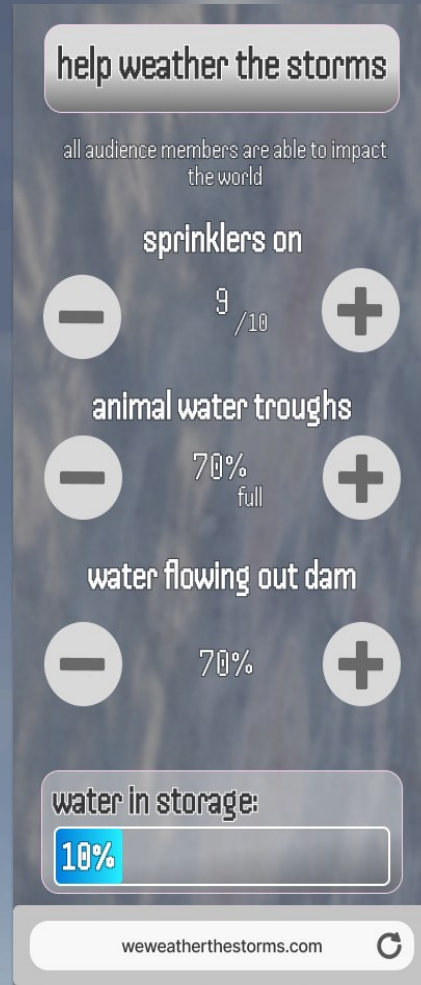
wireframes of the three interaction controllers

feedback provided
visually and through
soundscape

buttons can only be pressed every
5 seconds per device (apart from
'harvest all crops' which can
collectively be pressed every
thirty seconds)

simple UI with a lack of
personal notifications/
messages

invites audiences to reconsider
phones and reclaim agency over their
tech, challenging blackboxed software



each controller is aesthetically
different so players can
identify each other easily

no collection of personal
data

no personal levelling up /
tasks, each player is part
of the collective and does
not need to be maximised or
endlessly improved

utilises touchscreen
interactions players will be
used to doing

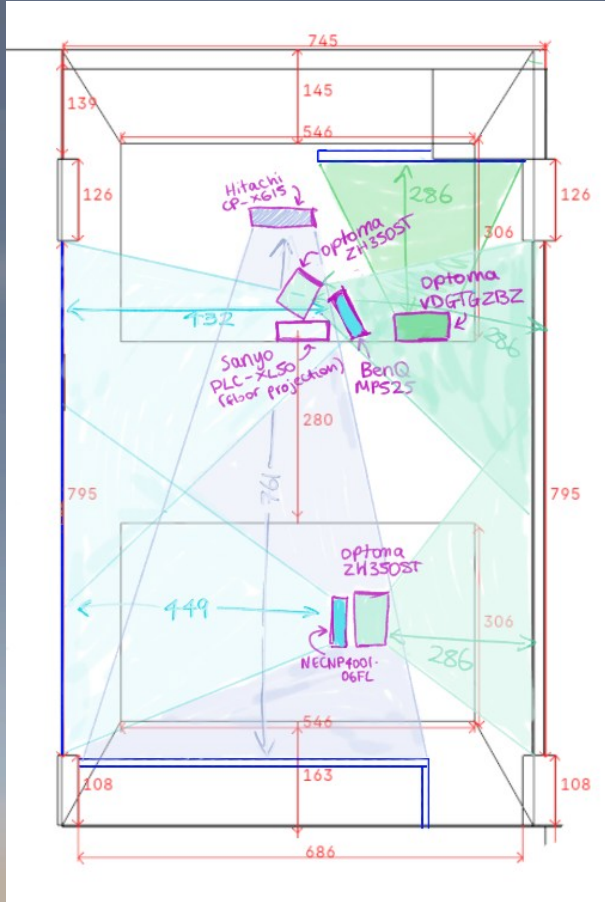
all actions are based on
real human actions

Using phones as controllers allows everyone the chance to interact with the space individually and become part of the collective group making impacts to the digital space. It looks to democratise interaction as no one person is singled out as the sole controller of the world, which could become a pressurised experience for the one player as their game interactions start to be watched by all other audience members.

I have built webpages
to communicate with Unity games before

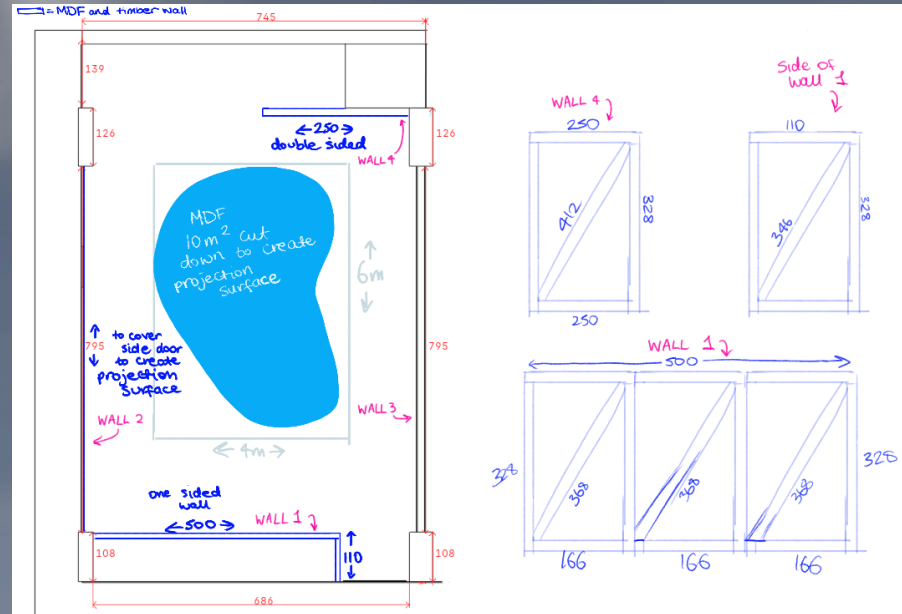
FIRST ROOM

immersive projection room



projector setup

walls built



Extra walls are built as projection surfaces within the room, so when audiences first walk in, they are immediately greeted with a projection of the world directly across from them. The additional wall to their left means they are not yet stood in the middle of the projection space, aiming to make first entrance to the space less overwhelming.

unity instances will communicate with each other using OSC commands so every screen across rooms shows the same multiplayer world, with phones communicating separately to these unity instances using websockets and a nodeJS server.



SECOND ROOM

The physical exhibition space is a playful, collaborative space grounded in the understanding that there can be no separation between technology and ecology, countering clean tech aesthetics. The wires, PCs, and other technological components required to show the digital ecosystem are not trying to be hidden or sanitised as a constant reminder to the audience of the materiality of the technology involved in the development and upkeep of the digital ecosystem. In this room, an abstract digital forest is built, and the individual organic technological forms become a physical manifestation of the tension that exists between building environmental games using digital technology and therefore increasing your carbon footprint, or even just using digital technology.



abstract forest room



tree displaying a clear view of the dam, reservoir and sunrise in the morning.

tree displaying the in-world time and weather forecast to allow players to plan ahead

tree displaying third person perspectives of different animals walking around the ecosystem

tree displaying green houses and plants growing within

< custom built benches >

tree displaying a wide view of the sunset on an evening

PUBLIC PROGRAM

DAILY GROUP PLAY SESSIONS

Due to the multiplayer aspect of this exhibition, audiences are encouraged to visit the exhibition at 1pm to be part of a larger group. This doesn't restrict people visiting at other times and doesn't need to be attended by the arebyte team as it just is guidance for visitors.



VIRTUAL NATURE WALK FOLLOWED BY DISCUSSION

Event providing the opportunity for players to stroll around the digital ecosystem in a first-person perspective, in a world that was not built to allow for this, revealing new perspectives across the immersive projections and screens. Encourages audiences to break the game, discover new perspectives, fall out of the world and understand the systems at play within the world, followed by a group discussion.

PLAYTESTING AND WORLDBUILDING WORKSHOPS

To allow worldbuilding to become a collaborative practice, I plan to open the worldbuilding and playtesting processes to the public through multiple workshops. This is to ensure the proposed 'utopian' future doesn't reflect the views and ambitions of a singular individual, and ensures systems embedded in code reflect diverse perspectives on climate justice. Importantly, this allows it to resist environmental colonial norms, acknowledging different world ontologies of environmental use and protection. To take place in arebyte's workshop space and/or online. I'm interested in working with community, school or university groups near to arebyte, particularly those involved in sustainability and urban rewilding.



BIO + ACKNOWLEDGEMENTS

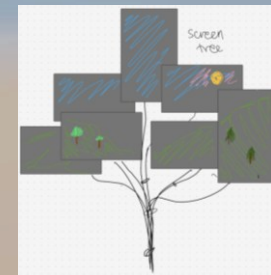
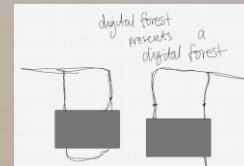
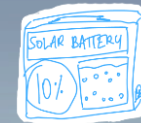
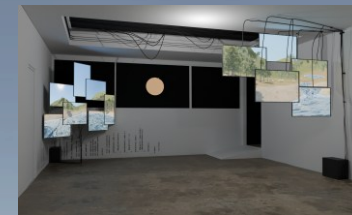


I am a digital artist and creative technologist from Leeds, UK, creating digital games, installations and immersive artworks with a socially engaged practice. My work is largely inspired by glitch art, solar punk, worlding, and video games, featuring forests, plants and environmental themes in a variety of contexts across time and space: worlds current, speculative and historic, frequently juxtaposing natural elements with the digital realm. I'm interested in the potential of digital experiences to educate and create explorative, playful and social spaces, particularly in environmental contexts, whilst leaning into game art and glitch aesthetics. My practice encapsulates game worlds and often includes realms that players are traditionally not supposed to see.

Thank you to my mentor Sam Mercer and all on the arebyte team (Pita, Laz, Claudel) for your support in developing this proposal.

find me at marytallontire.co.uk // [@marytallontire](https://twitter.com/marytallontire) // marytallontire02@outlook.com

sketches and renders from initial ideation



screenshots from game development

